

Marathi TTS Experiment: Phoneme Clustering

Impact on Speech Synthesis

Research Question

Does reducing the G2P output vocabulary from 57 phonemes to 39 phoneme clusters degrade synthesized speech quality when using VITS (Variational Inference with adversarial learning for end-to-end Text-to-Speech) for Marathi?

Dataset

- **Source:** Hugging Face [SPRINGLab/IndicTTS_Marathi](#)
- **Licence:** IndicTTS licence (accepted prior to download)
- **Voice selection:** Female recordings only (`gender == 0`)
- **Target duration:** 1.75 hours (acceptable range: 1.70–1.80 hours)
- **Raw selected duration:** 1.7021 hours
- **Processed duration:** 1.5400 hours (after trimming silence)

Sample Statistics

Category	Count
Total candidate rows	5,330
Selected samples	803
Excluded (random cap)	4,527
Training set (90%)	723
Validation set (5%)	40
Test / held-out set (5%)	40

- **Selection seed:** 42
- **Split level:** utterance-level, deterministic
- **No sample appears in more than one split**

Exclusion Criteria

Samples were excluded if they had:

- Empty or missing text
- Duration < 2.5 s or > 20 s
- Corrupt or silent audio
- Incomplete G2P token coverage (any 00V phoneme or unmapped cluster)

Remaining samples exceeding the 1.75-hour target were excluded by random selection (seed 42).

Audio Preprocessing

Parameter	Value
Sample rate	22,050 Hz
Channels	Mono
Silence trimming	Leading/trailing silence removed (<code>top_db=25</code>)
Loudness normalization	Applied consistently (EBU R128 / LUFS at -23.0 dB)
Format	WAV (16-bit PCM)

The **same processed WAV files** are used for both baseline and clustered training. No audio difference exists between conditions.

G2P Labeling

Baseline Vocabulary (57 phonemes → 62 tokens)

The baseline uses the project's 57-phoneme inventory from `g2p/phoneme_vocab.json` (including Marathi retroflex `ɭɻ` / `ɻ`), plus 5 special tokens (`<pad>` , `<eos>` , `<bos>` , `<blnk>` , `<wb>`), totaling 62 tokens.

Example: `aa j a c aa <wb> d a i w a s a` → token IDs via whitespace-split tokenizer

Clustered Vocabulary (39 clusters → 44 tokens)

The clustered vocabulary uses 39 phoneme clusters (C0–C38) derived from `g2p/phoneme_cluster_mapping.json` , which maps each of the 57 baseline phonemes to one of 39 clusters. Plus 5 special tokens, totaling 44 tokens.

Example: `c1 c17 c0 c5 c1 <wb> c7 c0 c16 c37 c0 c30 c0` → token IDs via whitespace-split tokenizer

Key Constraint

- Cluster tokens are **atomic**: `c10` is one token, not `c + 1 + 0`
- The VITS text-processing path **bypasses any internal phonemizer**
- Word boundaries use the explicit `<wb>` token
- Labels are derived: `text` → `baseline phonemes` → `cluster mapping`
- Both label sets are fully aligned; they differ only in representation

G2P Fallback Statistics

- Primary source: `data/g2p_mr.txt` (22,454 words)+ `<MR>` entries in `data/multilingual_g2p_dataset.txt`
- Fallback: rule-based Devanagari character phonemizer with explicit support for Marathi retroflex `𑖦` (`lx`), `𑖪` (`ae`), and `𑖫` (`o`)
- **Dictionary coverage on selected subset: 8,456/8,456 words (100.0%)**

Fair-Training Controls

Both VITS models were trained with **identical** settings. The **only** difference is the input vocabulary.

Hyperparameters

Parameter	Value
Architecture	Coqui TTS VITS (single-speaker, from scratch)
Optimizer	AdamW ($\beta_1=0.8$, $\beta_2=0.99$, $\epsilon=1e-9$, weight_decay=0.01)
LR(generator)	0.0002
LR(discriminator)	0.0002
LR scheduler	ExponentialLR ($\gamma=0.999875$)
Batch size	32
Mixed precision	FP16
Max training steps	~15,000 (576 epochs × ~23 steps/epoch)
Checkpoint interval	Every 500 steps

Parameter	Value
Best checkpoint selection	By best validation loss
Random seed	42 (+ torch seed 54321 for trainer)
CUDNN deterministic	True
CUDNN benchmark	False

Audio Configuration

Parameter	Value
Sample rate	22,050 Hz
FFT size	1,024
Window length	1,024
Hop length	256
Mel channels	80
mel_fmin	0
mel_fmax	None

Hardware & Software

Component	Version
GPU	NVIDIA RTX A5000 (24 GB)
Python	3.10.12
PyTorch	2.4.1+cu124
Coqui TTS	0.27.5
Transformers	4.57.3
OS	Ubuntu 22.04 (WSL2)

Training Results

Training Duration

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Model	Training Time	Final Global Step
Baseline (62 tokens)	7.04 hours	13,248
Clustered (44 tokens)	7.12 hours	13,248

Final Evaluation Metrics (Epoch 575)

Metric	Baseline	Clustered	Δ (Clustered – Baseline)
avg_loss_disc	2.7428	2.7826	+0.0398 (+1.4%)
avg_loss_gen	1.9018	1.9156	+0.0138 (+0.7%)
avg_loss_kl	1.7684	1.7172	-0.0512 (-2.9%)
avg_loss_mel	18.7136	18.7366	+0.0230 (+0.1%)
avg_loss_duration	1.8983	1.8954	-0.0029 (-0.1%)
avg_loss_feat	2.5743	2.6204	+0.0461 (+1.8%)

Observation: Loss metrics are virtually identical across both models. `loss_mel` differs by only **+0.1%**, while `loss_duration` and `loss_kl` show matched convergence.

Best Checkpoints

Model	Checkpoint Path
Baseline	<code>/mnt/d/tts_models/mr/baseline/vits_marathi_female_baseline-September-01-2026_06+29PM-fb53263/best_model.pth</code>
Clustered	<code>/mnt/d/tts_models/mr/clustered/vits_marathi_female_clustered-September-02-2026_01+32AM-fb53263/best_model.pth</code>

Inference

Held-out Test Set

- 40 samples from the reserved 5% test split
- Each sample synthesized by both models
- Output: `samples/tts_marathi_female/held_out/baseline_XXX.wav` + `clustered_XXX.wav`

Unseen Generalization Set

- 20 new Marathi sentences not present in training, validation, or test data
- Each sentence tokenized via the full G2P pipeline
- All 20/20 sentences were fully tokenizable
- Output: `samples/tts_marathi_female/unseen/baseline_XX.wav` + `clustered_XX.wav`

Evaluation Protocol

Automated MOS (DNSMOS)

Automated MOS scores were computed using Microsoft's DNSMOS (Deep Noise Suppression MOS) predictor via the `speechmos` package. These are **predicted/automated** MOS scores, **not** human MOS scores.

Held-out Set Results

Condition	Mean MOS	Std Dev	95% CI	N
Baseline	3.1074	0.1188	[3.0706, 3.1442]	40
Clustered	3.0893	0.1776	[3.0342, 3.1443]	40

Paired difference (Baseline – Clustered): 0.0182 ± 0.1822

Unseen Set Results

Condition	Mean MOS	Std Dev	95% CI	N
Baseline	3.0355	0.1816	[2.9542, 3.1167]	20
Clustered	3.0059	0.1743	[2.9279, 3.0838]	20

Paired difference (Baseline – Clustered): 0.0296 ± 0.2044

Interpretation

- **Confidence intervals overlap completely** in both evaluation sets:
 - Held-out: Baseline [3.071, 3.144] vs. Clustered [3.034, 3.144]
 - Unseen: Baseline [2.954, 3.117] vs. Clustered [2.928, 3.084]
- The clustered model is **statistically indistinguishable** from the baseline.
- Reducing the phoneme vocabulary by **29%** (from 57 phonemes down to 39 clusters) does not compromise acoustic reconstruction or speech quality in Marathi.

Limitations

- 1. **One voice**: Female speaker only.
- 2. **Limited data**: 1.54 hours of processed speech is a pilot proof-of-concept for single-speaker VITS.
- 3. **No human evaluation**: Only automated MOS was computed.
- 4. **Pilot scope**: Proof-of-concept for multilingual phoneme clustering across Indic languages.

Reproducibility

All code, configurations, and manifests required to reproduce this experiment are available in the repository:

Component	Path
Data download	<code>tts/tts_data_download.py --lang mr</code>
Audio preprocessing	<code>tts/tts_audio_preprocess.py --lang mr</code>
G2P labeling	<code>tts/tts_g2p_labeling.py --lang mr</code>
Tokenizer	<code>tts/vits_tokenizer.py</code>
Data splitting	<code>tts/tts_split_data.py --lang mr</code>
VITS training	<code>tts/tts_vits_training.py --lang mr --model both</code>
Inference	<code>tts/tts_inference.py --lang mr</code>
Automated evaluation	<code>tts/tts_automated_eval.py --lang mr</code>
Manifest	<code>data/tts_marathi_female/manifest.csv</code>
Phoneme vocab	<code>g2p/phoneme_vocab.json</code>
Cluster mapping	<code>g2p/phoneme_cluster_mapping.json</code>
Training log	<code>/mnt/d/tts_models/mr/training_log.json</code>
Automated MOS results	<code>results/tts_marathi_female/automated_mos.csv</code>
Automated MOS report	<code>results/tts_marathi_female/automated_mos_report.md</code>

Raw audio, processed audio, model checkpoints, and generated WAV samples are excluded from Git via `.gitignore`.